



- 8 (a) Find the values of k for which the system of equations

$$\begin{aligned} x - y + 2kz &= 1, \\ kx + y + 2z &= 2, \\ 2x - y + z &= 3, \end{aligned}$$

does not have a unique solution.

[3]

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- (b) Given that $k = -\frac{1}{2}$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically.

[3]

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- (c) Given instead that $k = -1$, show that the system of equations in part (a) is also inconsistent. Interpret this situation geometrically.

[4]

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The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{pmatrix}.$$

- (d) Use the characteristic equation of $\mathbf{A}$ to show that $\mathbf{A}^4 = p\mathbf{A}^2 + q\mathbf{A}$ where p and q are integers to be determined. [5]

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